Backend Mastery Timetable

TypeScript • Node • PostgreSQL • MongoDB • React • System Design

14 Oct 2026 !' 29 Jan 2027 | 16 weeks | 3 portfolio projects

1. Weekly Schedule

Mon & Tue: 4:00am - 7:00am Wed & Fri: 7:00am - 10:00am (12 hours per week, about 190 hours total)

How each 3-hour session is used:

- Hour 1: Theory (concept + why it exists + how it can be asked in an interview)
- Hours 2 to 2.5: You write the code. I review and correct. I don't write it for you.
- Last 30 min: Recall. Close everything and explain what you built out loud, from memory.

Ground rules

- You type all the code. I point out mistakes and explain why they are mistakes.
- Every feature: ask 'What is the time complexity?' and 'What happens at 1 million users?'
- Every project ends with a README, an architecture diagram and an oral defence with me as interviewer.
- Every Friday: one Big-O/DSA exercise and one mini system design question.
- Commit to GitHub after every session. Your commit history is proof that you built it.
- Wk 11 (Christmas) is a buffer week. Use it to catch up. Missed sessions slide into it.

2. Roadmap at a Glance

Block 1	Wk 1-5 | 14 Oct - 13 Nov	TS refresher, Node, Express, REST, PostgreSQL, Prisma, auth, React basics	**Project 1: Task & Notes Manager**
Block 2	Wk 6-11 | 16 Nov - 25 Dec	MongoDB, transactions, security, testing, Redis, queues, payments	**Project 2: E-commerce Platform (Postgres + Mongo)**
Block 3	Wk 12-16 | 28 Dec - 29 Jan	System design, WebSockets, Docker, CI/CD, scaling, deployment	**Project 3: Logistics Tracking Platform (capstone)**

BLOCK 1 | Foundations to Project 1

Week 1 | 14-16 Oct | TypeScript Refresher	
Wed 7-10am	Setup: Node, VS Code, Git, GitHub, tsconfig. Types, interfaces, type vs interface, unions, narrowing, enums. Mini task: typed functions.
Fri 7-10am	Generics, utility types (Partial, Pick, Omit, Record), classes, async/await typing. Build a typed CLI todo app. Big-O intro: O(1) vs O(n).

Weekly goal: Write typed TS comfortably without looking things up.

Week 2 | 19-23 Oct | HTTP, Node.js and Express	
Mon 4-7am	How the internet works: client/server, DNS, IP, ports. HTTP methods, status codes, headers, cookies. Big-O: O(n), O(log n).
Tue 4-7am	Node internals: event loop, call stack, promises, streams, fs, modules, env variables. Why Node is non-blocking.
Wed 7-10am	Express: routing, middleware, controllers/services structure, project layout with TS.
Fri 7-10am	REST design, Zod validation, global error middleware, custom errors. Build the CRUD API. Recall: explain a request's lifecycle.

Weekly goal: A working CRUD REST API (in-memory data).

Week 3 | 26-30 Oct | PostgreSQL and SQL	
Mon 4-7am	Relational model: tables, primary/foreign keys, 1:1, 1:N, M:N. Install Postgres, pgAdmin/psql. Design the Project 1 ERD.
Tue 4-7am	SQL: SELECT, WHERE, JOINs, GROUP BY, HAVING, aggregates, subqueries. Practice 20 queries
Wed 7-10am	Normalization, indexes (B-tree), EXPLAIN ANALYZE. Big-O of indexed vs full table scan.
Fri 7-10am	Prisma: schema, migrations, seeding, relations. Connect the API to the DB. Recall: why do indexes make reads faster and writes slower?

Weekly goal: Express API reading and writing to a real PostgreSQL database.

Week 4 | 2-6 Nov | Authentication and API Quality	
Mon 4-7am	Password hashing (bcrypt/argon2), sessions vs JWT, access + refresh tokens, httpOnly cookies.
Tue 4-7am	Auth middleware, protected routes, basic RBAC (user/admin), register/login/logout flow.
Wed 7-10am	Pagination, filtering, sorting, search, logging (pino), Helmet, CORS, rate limiting, Swagger docs.
Fri 7-10am	React + Vite + TypeScript: components, props, state, hooks, React Router. Big-O: hash maps.

Weekly goal: Secure API with login, roles, pagination, plus React project setup.

Week 5 | 9-13 Nov | PROJECT 1: Task & Notes Manager	
Mon 4-7am	TanStack Query for data fetching, React Hook Form + Zod, typed API client.
Tue 4-7am	Client auth flow: login page, token refresh, protected routes, error/loading states.
Wed 7-10am	Finish features: tags, search, pagination UI. Fix bugs. Deploy (Render/Railway + Vercel).
Fri 7-10am	README + architecture diagram. ORAL DEFENCE: explain the project end to end with no notes. Block 1 review.

Weekly goal: Project 1 deployed, documented and defended.

BLOCK 2 | Databases, Security and Project 2

Week 6 | 16-20 Nov | MongoDB and NoSQL	
Mon 4-7am	NoSQL concepts, documents vs rows, BSON, Mongo shell/Compass. Install Mongo (or Atlas).
Tue 4-7am	Mongoose: schemas, models, CRUD, validation, embedding vs referencing.
Wed 7-10am	Aggregation pipeline, indexes in Mongo, text search.
Fri 7-10am	Postgres vs Mongo trade-offs. Design Project 2: requirements, ERD (orders/users in Postgres, catalog/reviews in Mongo), API list.

Weekly goal: Know when to pick Postgres and when to pick MongoDB.

Week 7 | 23-27 Nov | Advanced SQL: Transactions and Performance	
Mon 4-7am	Transactions, ACID, isolation levels, rollback.
Tue 4-7am	Race conditions, row locks, optimistic vs pessimistic locking. Reproduce and fix an overselling-stock bug.
Wed 7-10am	N+1 problem, connection pooling, query optimisation. Big-O: sorting and binary search.
Fri 7-10am	Build catalogue (Mongo) and cart modules. Mini system design: URL shortener.

Weekly goal: Product catalogue and cart modules working; no race conditions.

Week 8 | 30 Nov - 4 Dec | Security and Testing	
Mon 4-7am	OWASP Top 10: SQL injection, XSS, CSRF, IDOR, broken auth. Attack and fix your own API.
Tue 4-7am	Full RBAC (customer/seller/admin), OAuth2 Google login, secrets management.
Wed 7-10am	Testing: Vitest, Supertest, test database, mocking. Write tests for auth and cart.
Fri 7-10am	Order module using DB transactions + tests. Big-O: stacks and queues.

Weekly goal: Orders module with transactions and tests passing.

Week 9 | 7-11 Dec | Redis, Queues and Payments	
Mon 4-7am	Redis: data types, TTL, cache-aside, cache invalidation. Cache the product list.
Tue 4-7am	Background jobs with BullMQ: email on order, retries, cron jobs.
Wed 7-10am	Paystack/Stripe: checkout, webhooks, idempotency, signature verification.
Fri 7-10am	File uploads (Cloudinary/S3), Swagger docs. Start admin dashboard in React.

Weekly goal: Cached catalogue, background emails, working payment flow.

Week 10 | 14-18 Dec | PROJECT 2: E-commerce Platform	
Mon 4-7am	Admin dashboard: product CRUD, orders table, basic charts.
Tue 4-7am	Customer storefront: product list, cart, checkout, order history.
Wed 7-10am	Integration tests, bug fixing, Docker file draft, deploy.
Fri 7-10am	README + architecture diagram. ORAL DEFENCE. Mini system design: notification system.

Weekly goal: Project 2 live, documented and defended.

Week 11 | 21-25 Dec | Buffer Week (Christmas) and Revision	
Mon 4-7am	Revise weakest topics (list from defences). Redo missed exercises.
Tue 4-7am	DSA: trees, graphs (BFS/DFS), heaps. Solve 5 problems.
Wed 7-10am	Docker basics: images, containers, volumes, docker-compose with Postgres, Mongo, Redis.
Fri 7-10am	Optional / rest. Light review of Block 2 notes.

Weekly goal: Fill every gap from Blocks 1 and 2. Rest if you are on track.

BLOCK 3 | System Design and Capstone

Week 12 | 28 Dec - 1 Jan | System Design Method and Project 3 Kickoff	
Mon 4-7am	System design framework: requirements, estimation (QPS, storage, bandwidth), API, data model, scaling, trade-offs.
Tue 4-7am	Design the Logistics Platform: components, ERD, API, state diagram for shipments.
Wed 7-10am	WebSockets and Socket.io: rooms, events, auth on sockets, scaling sockets.
Fri 7-10am	Bootstrap Project 3 (1 Jan is a holiday, optional shorter session): repo, auth, users/drivers/ shipments modules.

Weekly goal: Design doc draft + project skeleton (monorepo, auth, shipments).

Week 13 | 4-8 Jan | Real-time and Core Logistics Features	
Mon 4-7am	Shipment lifecycle as a state machine (created, assigned, in transit, delivered). Validation of transitions.
Tue 4-7am	Driver assignment algorithm. Big-O: priority queue/heap, nearest driver search, geo indexes.
Wed 7-10am	Real-time tracking: Socket.io + Redis pub/sub, location updates.
Fri 7-10am	Notifications through queue (email/SMS), webhooks. Mini system design: chat app.

Weekly goal: Live shipment tracking working end to end.

Week 14 | 11-15 Jan | DevOps, Performance and Frontend	
Mon 4-7am	Dockerise app: multi-stage Dockerfile, docker-compose for all services.
Tue 4-7am	CI/CD with GitHub Actions: lint, test, build, deploy.
Wed 7-10am	React tracking UI: map, live updates, admin and driver views.
Fri 7-10am	Caching, rate limiting, monitoring, Sentry, health checks. Load test with autocannon/k6.

Weekly goal: Dockerised app with CI pipeline and live tracking UI.

Week 15 | 18-22 Jan | Scaling and Deployment	
Mon 4-7am	Monolith vs modular monolith vs microservices. Load balancers, horizontal vs vertical scaling.
Tue 4-7am	CAP theorem, replication, sharding, CDN, API gateway. Apply these to the logistics design.
Wed 7-10am	Hardening: tests, error handling, security review, backups and migrations.
Fri 7-10am	Deploy: Render/Railway or VPS with Nginx, HTTPS, env management.

Weekly goal: Capstone deployed and stable.

Week 16 | 25-29 Jan | PROJECT 3: Final Defence	
Mon 4-7am	System design document: diagrams, trade-offs, scaling plan. Final README.
Tue 4-7am	Mock interview 1: system design (design a delivery tracking app from scratch).
Wed 7-10am	Mock interview 2: explain your three projects cold, plus Big-O and database questions.
Fri 7-10am	Final presentation. Clean GitHub, CV, LinkedIn, portfolio site. Plan the job-hunt.

Weekly goal: Explain your system with no material, at interview level.

3. The Three Projects

Project 1: Task & Notes Manager (end of Wk 5)

- Register/login with JWT access + refresh tokens
- Tasks, notes, tags, search, pagination, filtering
- PostgreSQL + Prisma, Zod validation, Swagger docs
- React + TS frontend with protected routes
- You can explain: request lifecycle, auth flow, indexes, REST design

Project 2: E-commerce Platform (end of Wk 10)

- Products and reviews in MongoDB; users, orders and payments in PostgreSQL
- Roles: customer, seller, admin; Google login
- Transactional checkout with no overselling
- Redis caching, BullMQ emails, Paystack/Stripe with webhooks
- Tests with Vitest/Supertest; React storefront + admin dashboard
- You can explain: transactions, race conditions, caching, queues, security

Project 3: Logistics Tracking Platform (end of Wk 16)

- Customers, drivers, admins; shipment lifecycle state machine
- Driver assignment, live tracking with Socket.io + Redis pub/sub
- Queued notifications, webhooks, rate limiting, monitoring
- Docker, CI/CD, deployed live
- A written system design document with scaling plan
- You can explain: the entire architecture, trade-offs, and how it would handle 1M users

4. Progress Checklist (tick as you finish)

- [ ] Wk 1: TS refresher done
- [ ] Wk 5: Project 1 defended
- [ ] Wk 6: Mongo basics done
- [ ] Wk 8: Own API hacked and fixed
- [ ] Wk 10: Project 2 defended
- [ ] Wk 12: System design doc drafted
- [ ] Wk 14: App Dockerised + CI
- [ ] Wk 16: Project 3 defended, mock interviews done

Rule: the day you stop understanding why, you stop typing. Ask me first.